
M.Tech. Major-2 Project

RTL-Based Digital Control Architecture for Infrared Imaging System ROIC

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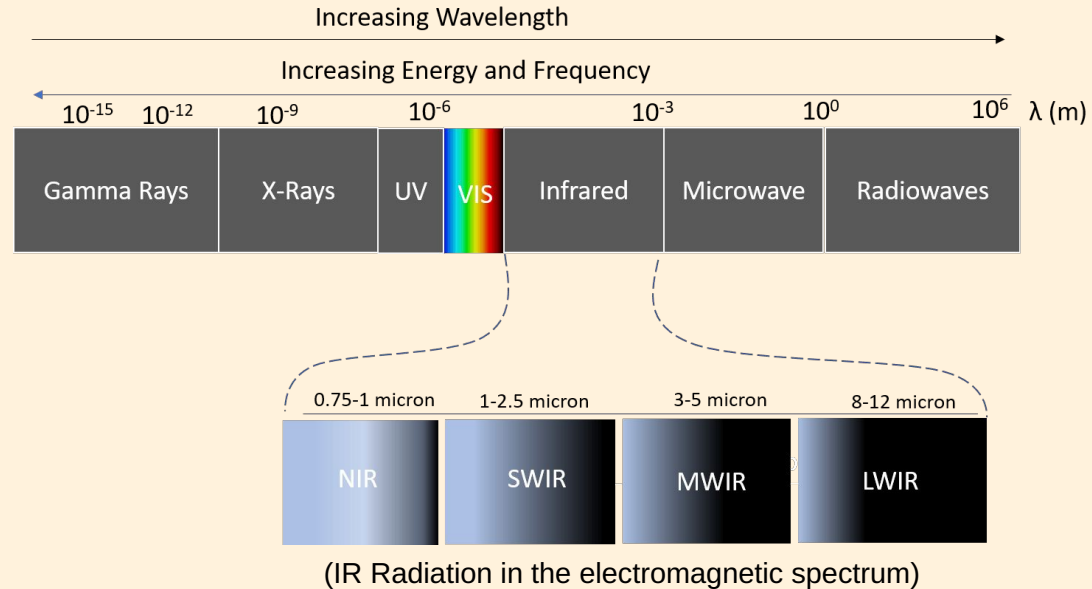
DRDO, New Delhi

Presentation Overview

- ❖ Introduction (Infrared Radiation)
- ❖ ROIC Concept Overview
 - IR Imaging System
 - Cell Architectures
- ❖ VLSI Design Flow
- ❖ Digital Design
 - Architecture, Features
 - Simulation Results
- ❖ Physical Design
- ❖ Conclusion & Future Work
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Introduction : Infrared Radiation

- ❑ Electromagnetic radiation with wavelengths longer than visible light, but shorter than microwaves.
- ❑ Emitted by all objects above absolute zero; intensity increases with temperature
- ❑ Easily absorbed by many materials, causing molecular vibration and heat.
- ❑ Different bands correspond to different sensing mechanisms and applications

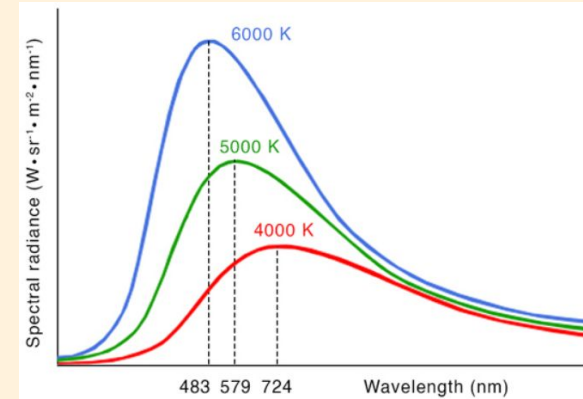


Thermal Radiation and Blackbody Behavior

- ❑ All objects above 0 K emit thermal radiation
- ❑ Ideal emitter: Blackbody (emits maximum possible radiation)

Key Laws : Establishing relations between important parameters

- ❑ **Planck's Law** ($E \propto \text{freq}$): Higher frequency light carries more energy in its individual photons
- ❑ **Stefan-Boltzmann Law** ($P \propto T^4$): Power per unit surface area radiated by a black body is directly proportional to the fourth power of its absolute thermodynamic temperature.
- ❑ **Wien's Displacement Law** ($\lambda_{\text{max}} \propto 1/T$): Higher temp objects emit radiation at shorter wavelength.
- ❑ **Blackbody Radiation** : Continuous spectrum of electromagnetic energy emitted by an ideal, opaque object that absorbs all incident radiation. This thermal radiation depends solely on the object's temperature, not its material.



IR Emission vs Reflection

- ❑ **Emissivity** : Determines how efficiently an object emits radiation
- ❑ Real objects are not ideal blackbodies, so emissivity is less than 1
- ❑ Affects accuracy of IR sensing
- ❑ Objects can:
 - ❑ **Emit IR radiation** : Human, animal, car engine, jet engine
 - ❑ **Reflect IR radiation** : Polished metal (reflect strongly)
 - ❑ Reflected radiation is not due to object temperature and depicts surrounding temperature

Main Thermal Imaging Bands :

- ❑ **MWIR (3-5 μm)** : Hot combustion objects like hot engines, missile plumes, recently fired weapons
- ❑ **LWIR (8-14)** : Normal temperature objects like human, animals, buildings, terrain, vehicles

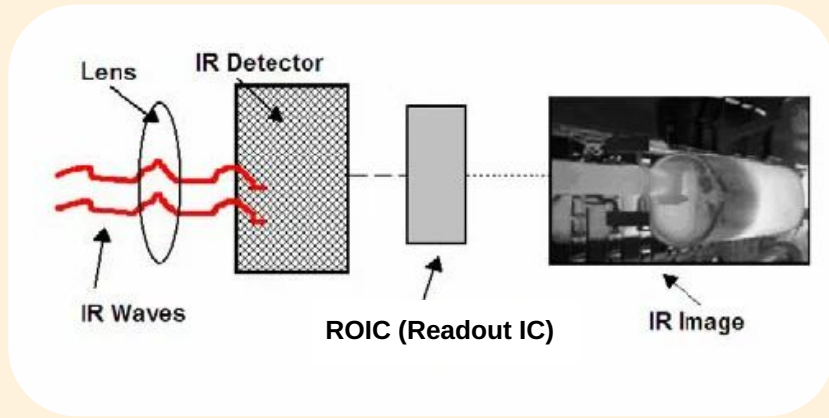
Design Flow Overview

ROIC
Concept
Overview

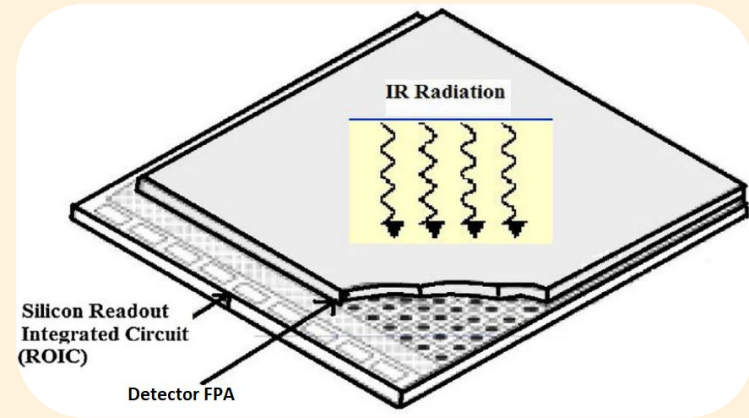
Digital
Design

Physical
Design

IR Imaging System



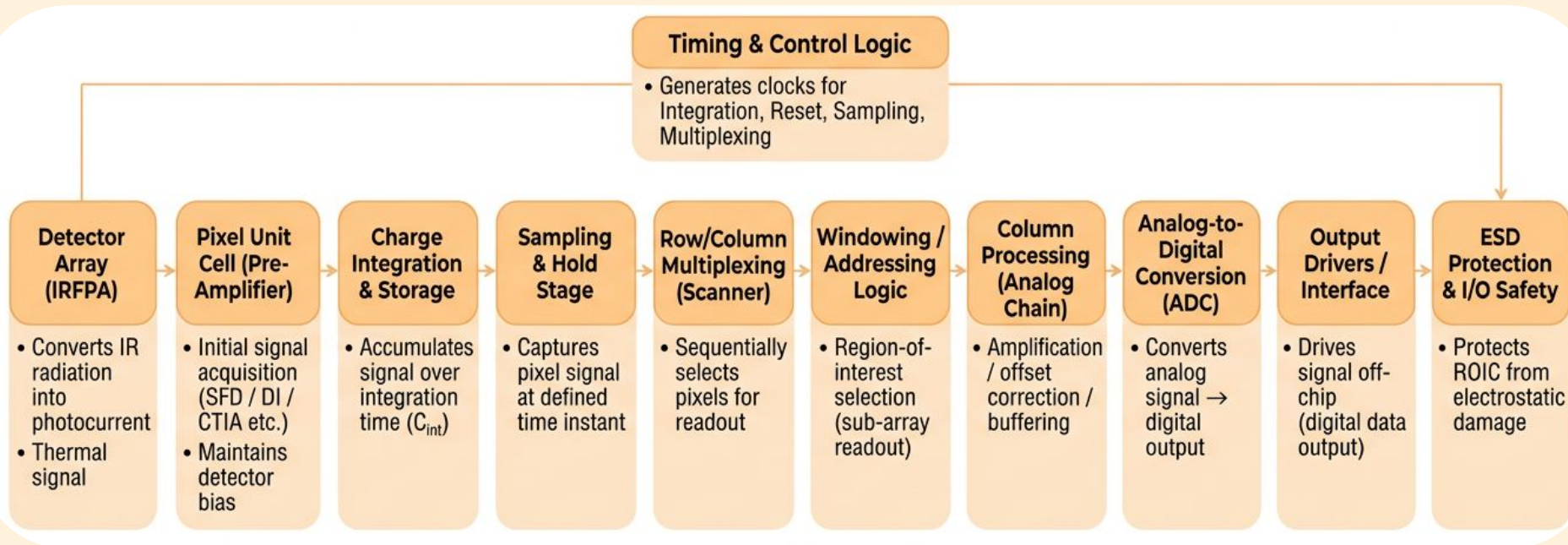
(Basic Components of IR Imaging Device)



(Two-Chip Hybrid System)

- ❖ Detector Chip (Top Layer):
 - HgCdTe (MCT) → tunable bandgap (SWIR–LWIR)
 - InSb → high sensitivity (MWIR)
 - Converts IR radiation → photocurrent / temperature change
- ❖ ROIC (Bottom Layer):
 - CMOS-based readout circuit
 - Current-to-voltage conversion, integration, multiplexing
- ❖ Indium Bump Bonding : Pixel-to-pixel high density interconnection

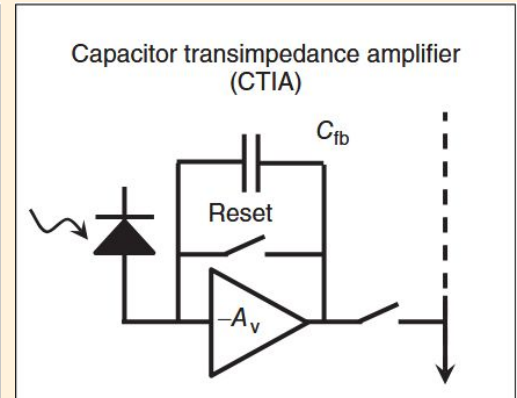
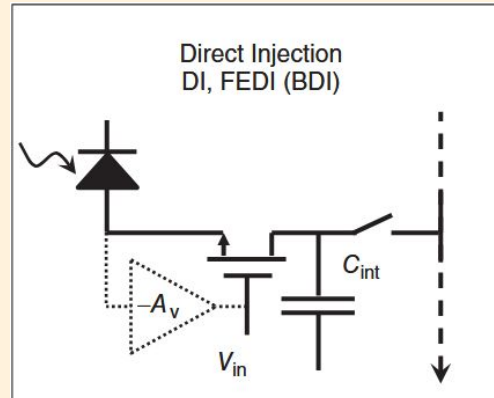
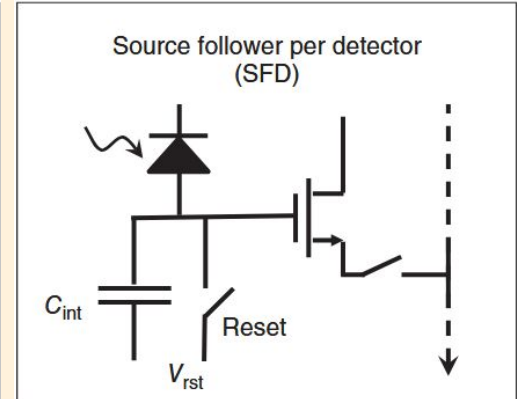
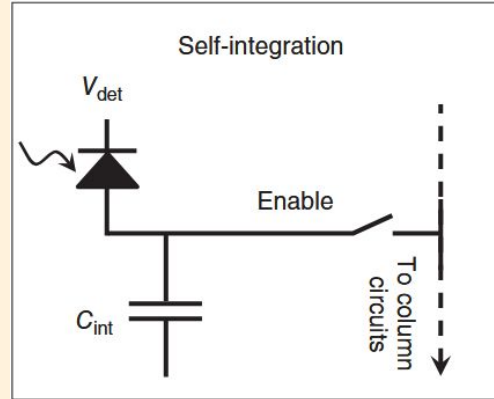
ROIC Functions



CMOS ROIC : Architecture and Evolution

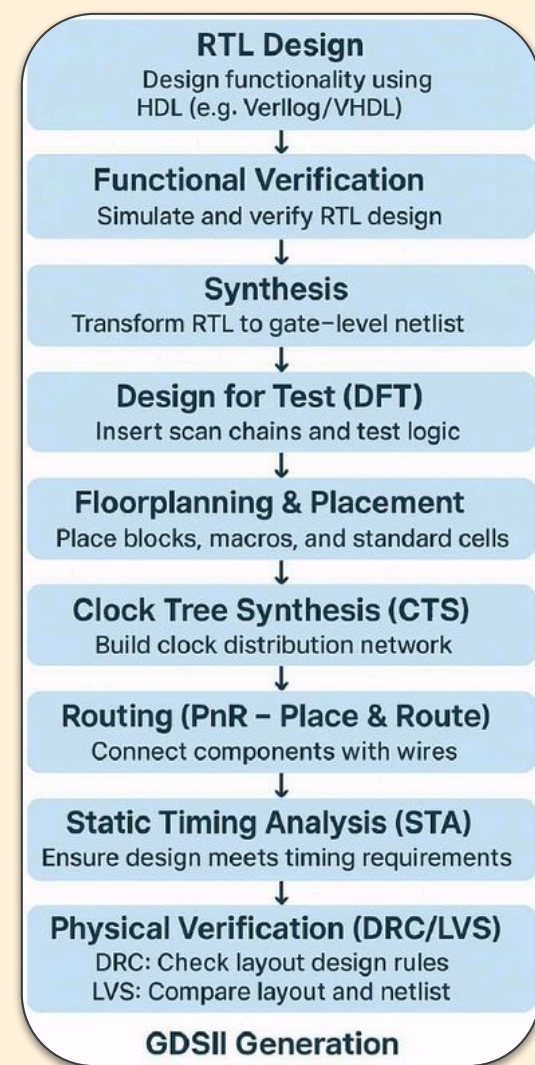
Key Design Parameters:

- ❑ Detector Bias Stability
- ❑ Linearity
- ❑ Injection Efficiency
- ❑ Sensitivity
- ❑ Noise
- ❑ Pixel Area
- ❑ Power Consumption



VLSI Design Flow (RTL to GDSII)

- ❑ RTL design and verification in **AMD Xilinx Vivado**
- ❑ Logic Synthesis (Gate-Level Netlist) in **Cadence Genus**
- ❑ Physical implementation in **Cadence Innovus**
- ❑ Timing/physical checks followed by **GDSII generation**



Design Flow Overview

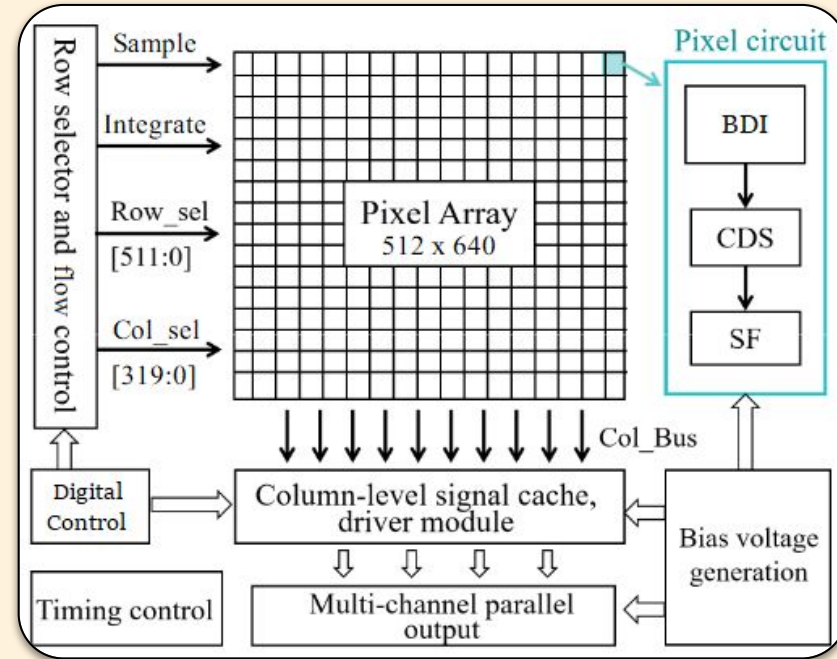
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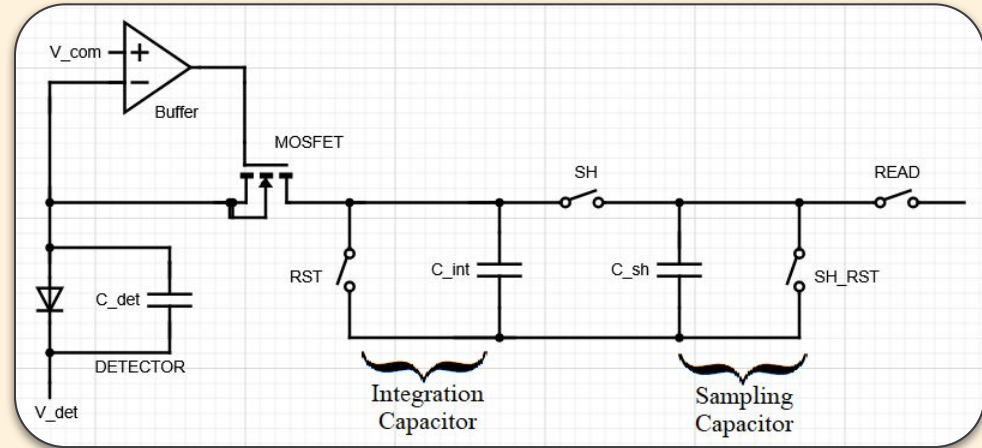
Digital ROIC Design Overview

- ❑ The BDI Pixel array circuit, bias voltage generation, output buffers, multiplexers are analog circuit designed and synthesised via custom ASIC design flow.
- ❑ The digital block supervises the readout operation of the ROIC and coordinates the interaction between the pixel array, analog readout path, and external timing inputs.
- ❑ It ensures that pixel signals are accessed in an ordered and synchronized manner for proper image formation.



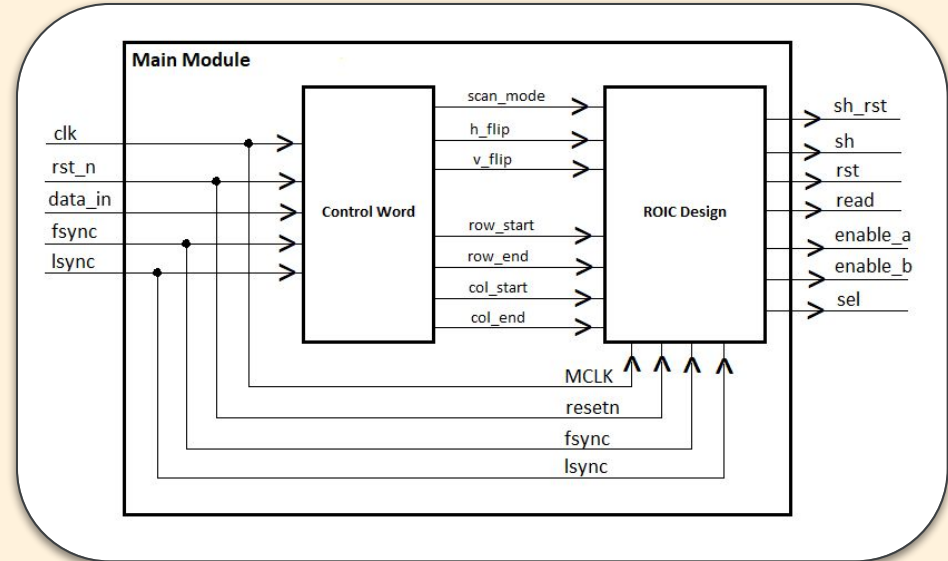
ROIC Cell : Signal Flow

- ❑ The detector current is first integrated on the integration capacitor and then transferred through the sample-and-read path under digital switch control.
- ❑ Control signals such as **RST**, **SH_RST**, **SH**, and **READ** define the sequence of reset, integration, sample, and output transfer operations.
- ❑ The switching sequence ensures that charge integration, sampling, and readout occur in the required order for stable signal acquisition.



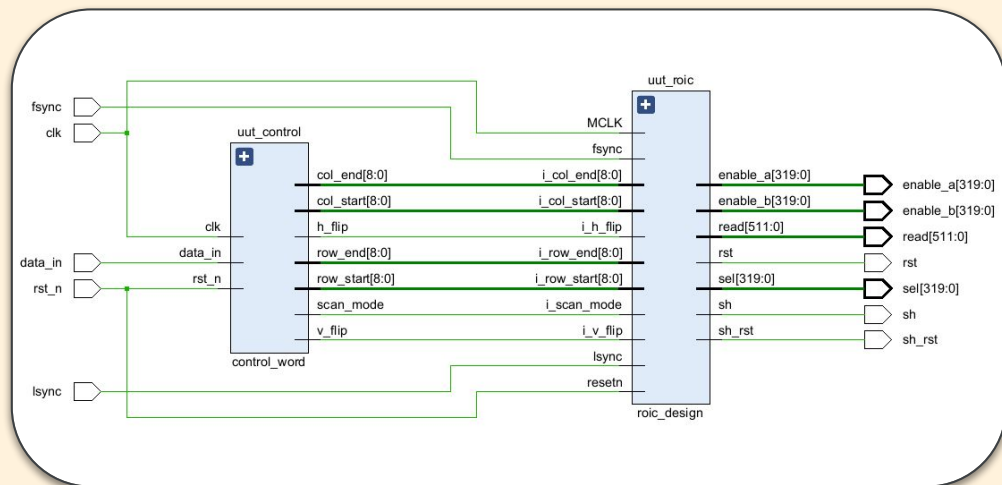
Top Level RTL Architecture

- ❑ The RTL architecture is organized around a top module that connects the **control word block** with the **ROIC design block**.
- ❑ The control word section captures frame-wise programmable inputs such as scan mode, readout direction, and row/column window boundaries
- ❑ The ROIC design block uses these inputs together with **CLK**, **FSYNC**, **LSYNC**, and reset to generate the required outputs.



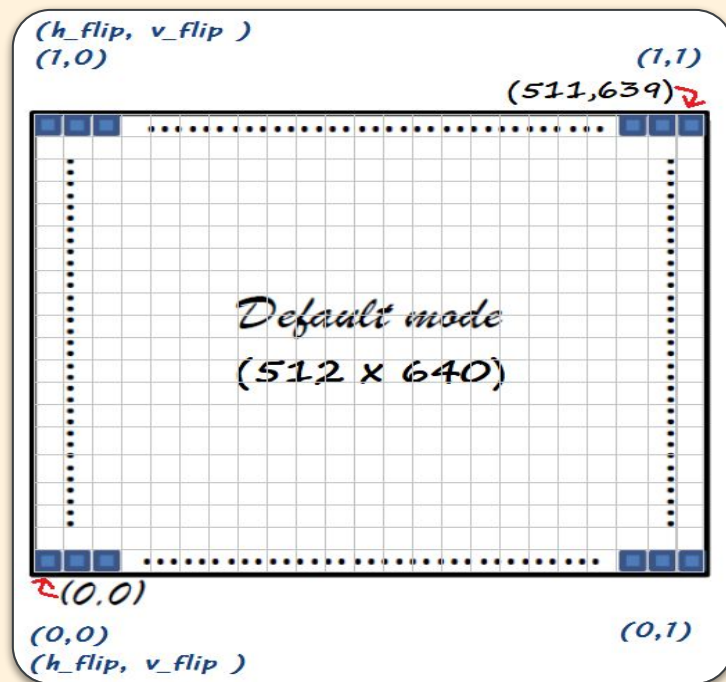
RTL Lint / Structural Realization

- ❑ The generated structural view matches the intended top-level architecture, with the control block connected to the ROIC logic through the expected programmable and timing signals.
- ❑ This step verified that the coded RTL preserved the planned hierarchy and signal flow before synthesis.



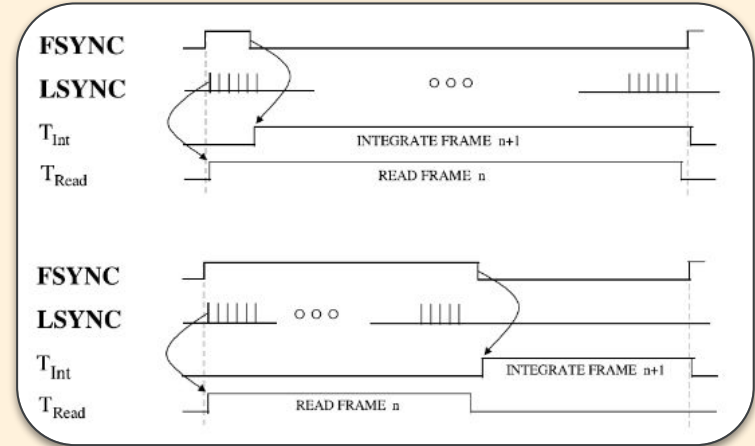
Readout Control Features

- ❑ The controller supports configurable scan direction through **hflip** and **vflip**, allowing the array traversal order to be modified at runtime.
- ❑ This makes the digital block flexible for different image orientations without changing the underlying array structure.
- ❑ It's useful when the required output order must match system-level image alignment requirements



Feature 2: Scan Modes ITR IWR

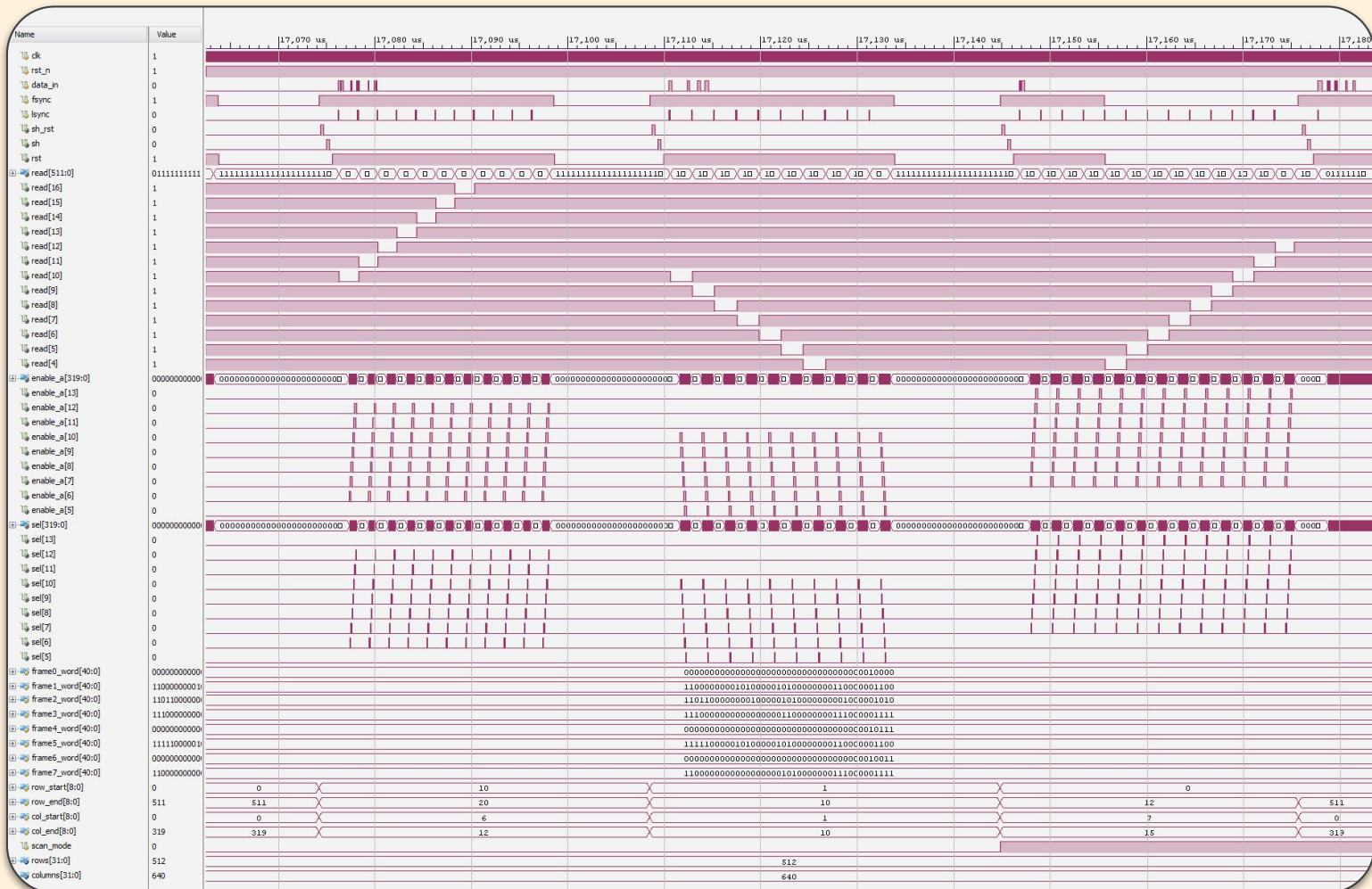
- ❑ The digital controller supports two scan modes: **ITR** and **IWR**, selected through the programmable control word.
- ❑ In **ITR**, integration and readout occur in separate intervals, which provides a clear timing separation between exposure and frame readout.
- ❑ In **IWR**, integration of the next frame overlaps with readout of the current frame, improving temporal efficiency and enabling faster operation.
- ❑ This mode flexibility allows the same digital architecture to support different imaging and timing requirements



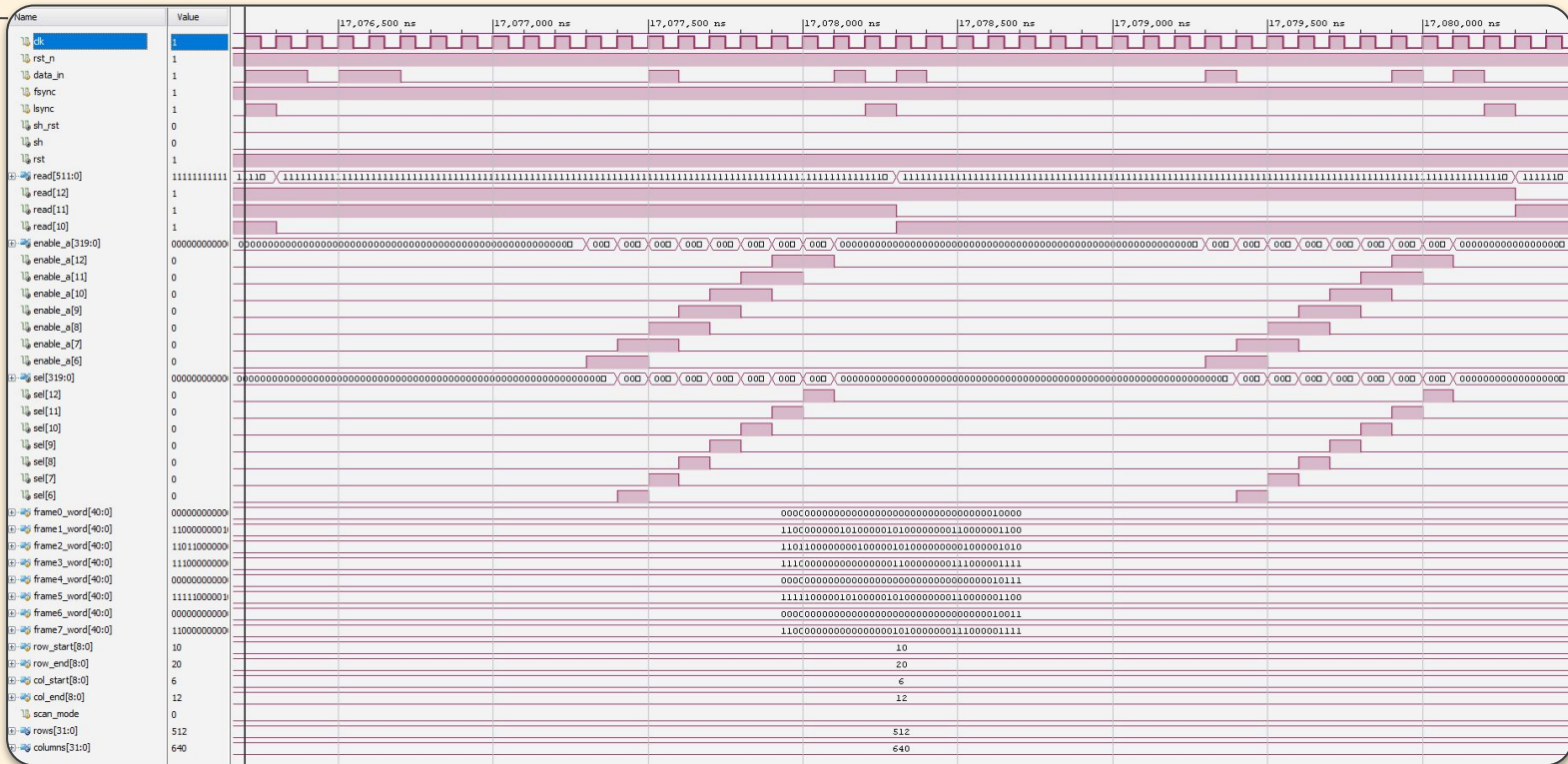
Feature 3: Windowing

- ❑ Windowing allows the controller to read only a selected subset of rows and columns instead of scanning the full 512×640 array
- ❑ The control word adjusts the active row and column boundaries by programming **row_start**, **row_end**, **col_start**, and **col_end**.
- ❑ As the selected window becomes smaller, the effective readout length is reduced, which lowers total frame readout time and increases achievable frame rate.
- ❑ This makes it possible to focus on a region of interest and acquire relevant image data more quickly.

Results



ROIC Simulation Result I



ROIC Simulation Result (Zoomed in : row)

Results

- ❑ The timing report shows zero negative slack and positive setup/hold margins, confirming that all signals meet the required timing and arrive at destination within the allowed clock period

Design Timing Summary

Setup

Worst Negative Slack (WNS): 88.420 ns

Total Negative Slack (TNS): 0.000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 4962

Hold

Worst Hold Slack (WHS): 0.137 ns

Total Hold Slack (THS): 0.000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 4962

Pulse Width

Worst Pulse Width Slack (WPWS): 49.500 ns

Total Pulse Width Negative Slack (TPWS): 0.000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 1346

All user specified timing constraints are met.

ROIC Simulation Result : Timing Analysis

RTL Synthesis & DFT(Design for Testing)

- ❑ Verified RTL of the digital ROIC controller was synthesized in Cadence Genus.
- ❑ The RTL description was translated into a **gate-level netlist** using standard-cell mapping and design constraints.
- ❑ **Scan chain insertion** was carried out as part of the DFT preparation to improve testability of the sequential logic.
- ❑ The resulting **final synthesized netlist** was then forwarded to the backend physical design flow.

Design Flow Overview

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Physical Implementation of the Digital Block

- After synthesis, the digital controller was taken through the backend implementation flow using Cadence tools.
- The objective was to convert the synthesized RTL into a layout-ready and physically realizable digital block.
- The main implementation stages included floorplanning, power planning, placement, clock tree synthesis, and routing.
- This flow established the physical structure required for later integration with the complete ROIC system.

Floorplan and Power Planning

- ❑ Floorplanning was first carried out to define the core region, placement boundaries, and overall physical organization of the digital block.
- ❑ This stage established the spatial framework for standard-cell implementation and interconnect distribution.
- ❑ Power planning was then performed to create the supply and ground network required across the block.

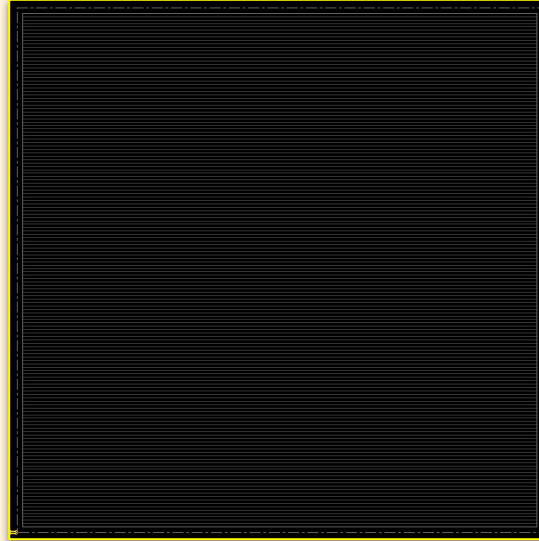


Fig: Floorplan

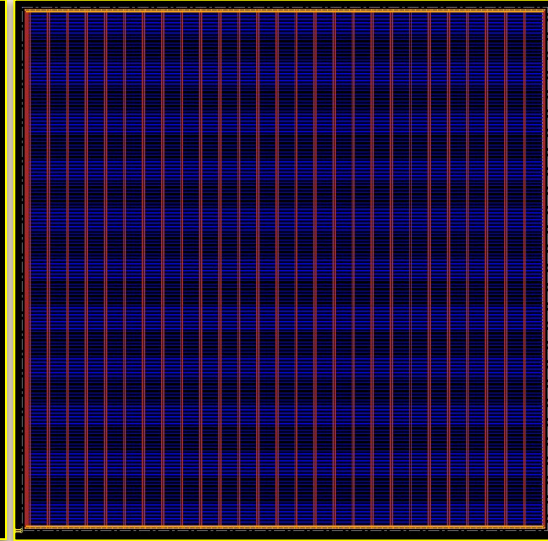


Fig: Power Plan

Placement

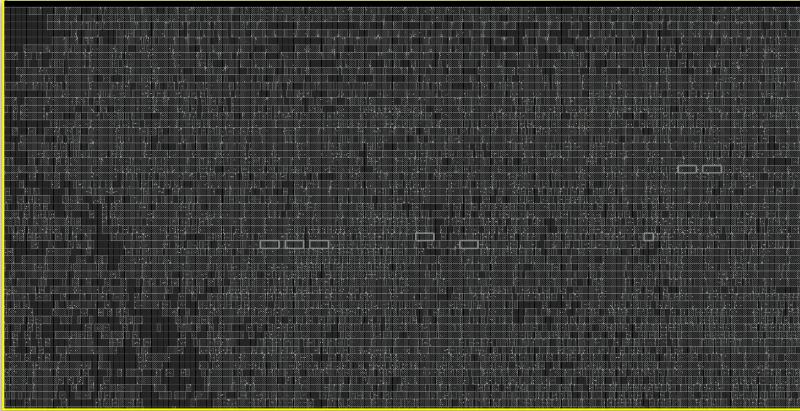
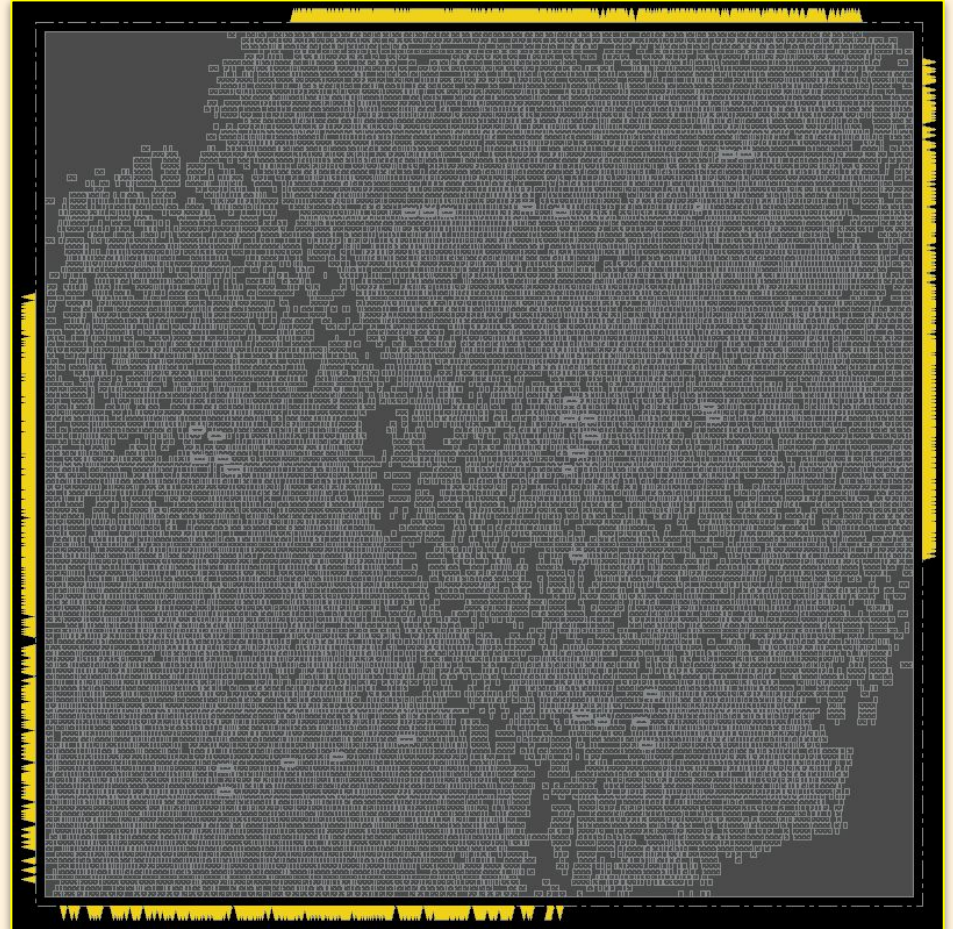
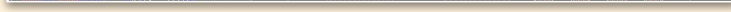


Fig: Std Cells Zoomed in

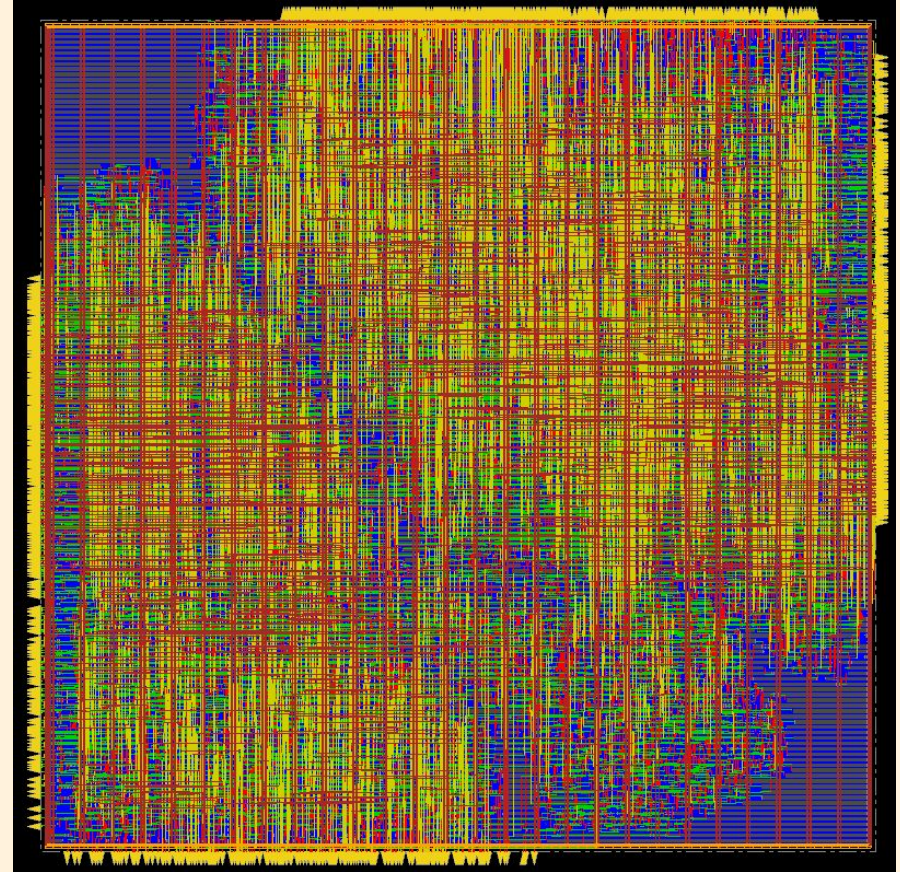
- ❑ Standard-cell placement was performed to arrange the synthesized logic cells in an organized and implementation-ready manner.





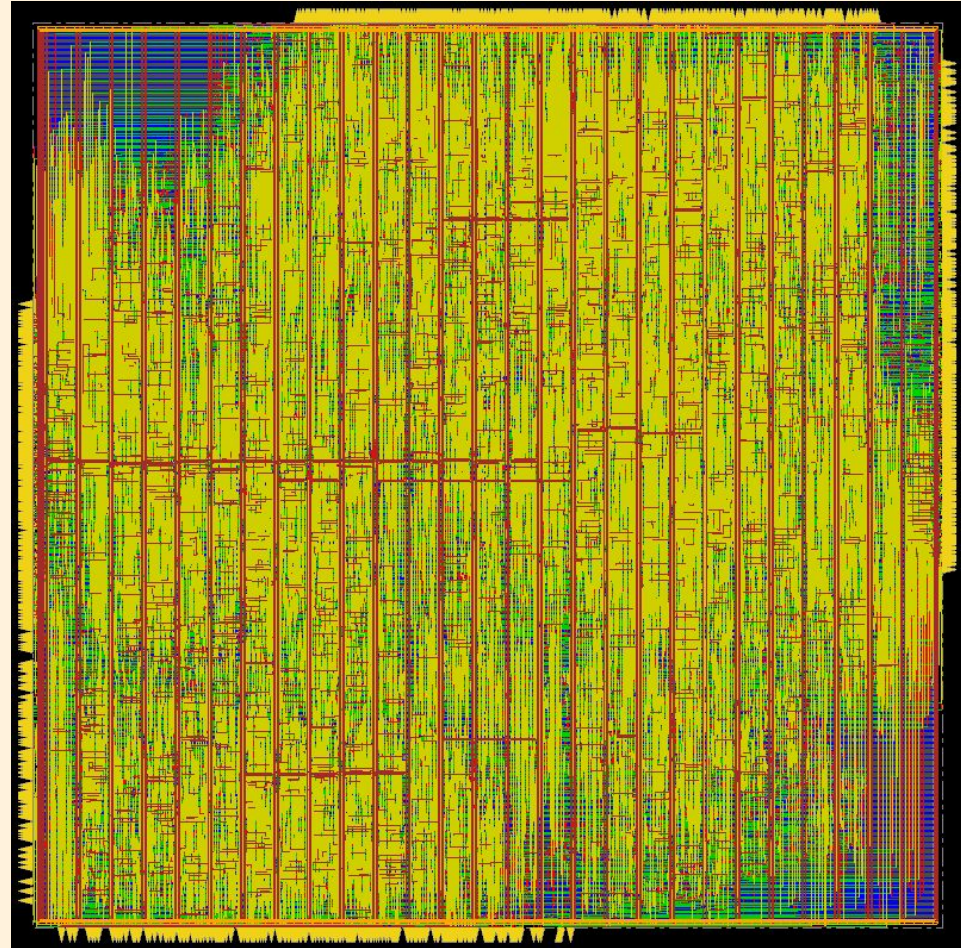
Clock Tree Synthesis & Routing

- ❑ Clock tree synthesis was used to distribute the clock signal across the design with controlled skew and balanced timing.
- ❑ After CTS, routing completed the interconnection of placed cells and clock networks.
- ❑ These stages converted the logical design into a connected physical implementation of the digital controller.



Physical Design : Layout

- ❑ The backend flow resulted in a clean physical realization of the digital ROIC controller.
- ❑ The design progressed from synthesized RTL to an implemented layout through standard physical design stages.
- ❑ This forms the digital foundation for integration with the larger mixed-signal ROIC architecture.



Conclusion & Future Work

Conclusion

- ❑ Developed and verified the RTL-based digital control architecture of the ROIC, supporting timing control, programmable readout order, windowing, and IWR/ITR operation; the design was also taken through the Cadence RTL-to-GDSII flow with timing constraints satisfied and implementation feasibility confirmed.

Future Work

- ❑ Extend the work toward full mixed-signal top-level integration by interfacing the digital controller with the analog ROIC blocks, completing pad ring and pin planning, and preparing the final integrated GDS for fabrication-ready tapeout.

References

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Thank You!